Contents

Volume 2 - Applications Guide: Appendix C - Strategy Analysis Case Studies

Appendix C1 - Strategic Analysis Case Study 1

1 Long Term Budget Forecasts and Performance Trends

This case study demonstrates the use of HDM-4 in budget forecasting. The objective of this study is to determine the required funding levels for user defined network performance standards, and to demonstrate the effect of budgetary constraints on the long term performance trends. This involves defining the road network in terms of representative sections and assigning alternative maintenance standards (investment alternatives) to each section.

The following steps are followed:

1 **Locate the case study data** (see Section 1.1)

2 **Review the case study data** (see Section 1.2)

3 **Run strategy analysis (without budget constraint)** (see Section 1.3)

4 **Perform budget optimisation** (see Section 1.4)

1.1 Locate the case study data

The case study data are included in the default database installed with the HDM-4 software. The data for this case study are located in the **Strategies folder** in the Case Studies Workspace**.** The name of the case study is **1: Long term budget forecasts and performance trends**.

To locate the data for this case study:

- Go to the Case Studies Workspace
- Open the **Strategies folder**

Double-click on the case study named:

1: **Long term budget forecasts and performance trends**.

1.2 Review the case study input data

The data can be reviewed under the HDM-4 Strategies work flow buttons and associated Tabs listed below:

- **Define Strategy Details** (see Section 1.2.1)

  The following screens may be displayed:

  - General
  - Select Sections
  - Select Vehicles

- ❑ Define Normal Traffic

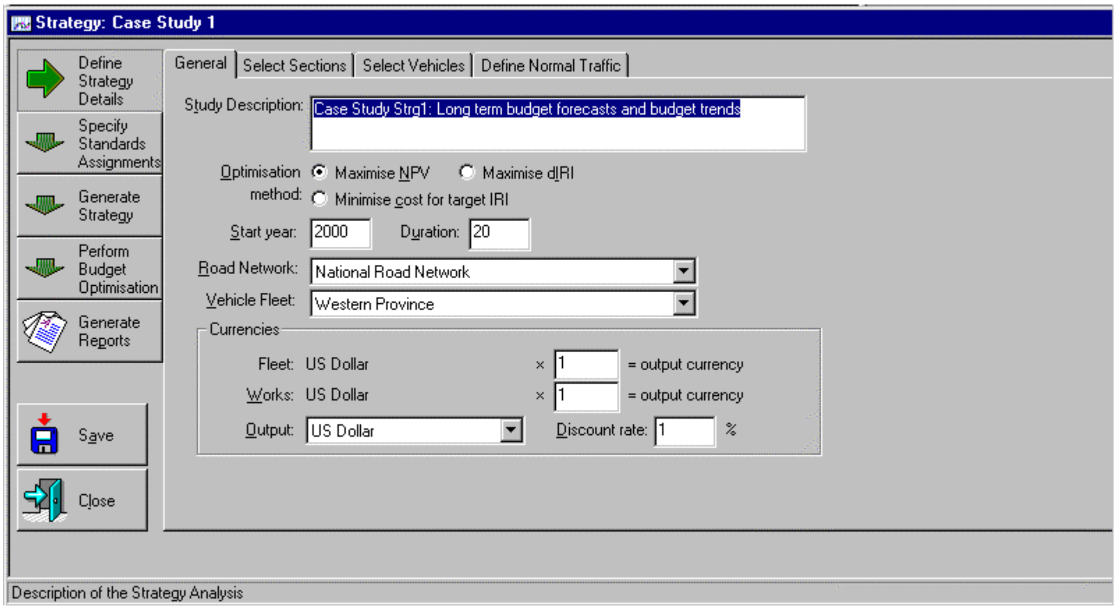
- ■ **Specify Standards Assignments** (see Section 1.2.2)

  The following screen may be displayed:

  - ❑ Alternatives

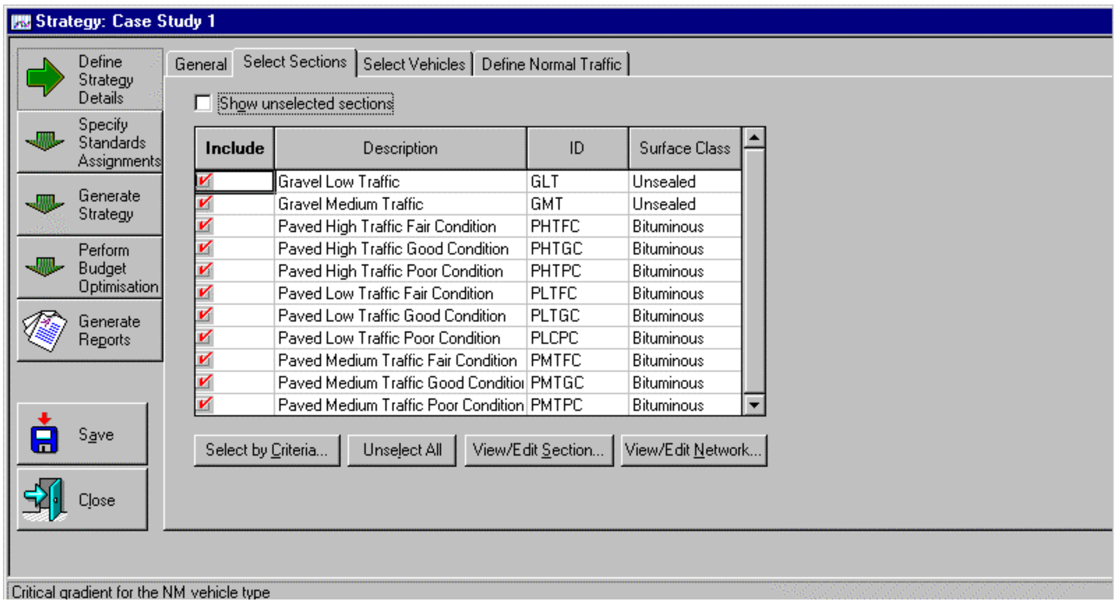
1.2.1 Define Strategy Details

General



This screen confirms the study description, optimisation method, analysis period (start year and duration) and the names of the pre-defined Road Network and Vehicle Fleet. For this case study the selected optimisation method is **maximisation of benefits (NPV)**.

Select Sections



The list of representative sections is displayed under the **Select Sections** Tab.

The road network under study has been modelled by eleven representative sections. The representative sections for bituminous (paved) roads have been based on traffic volume (**High**, **Medium** or **Low**) and road condition (**Good**, **Fair** and **Poor**), giving a total of nine sections.

The section ID (5 letters) for the bituminous sections is derived as shown by the following example:

For PHTGC:

P	Paved
HT	High Traffic
GC	Good Condition

Unsealed (gravel) roads have been classified by traffic volume alone, since condition may change rapidly in any given year. For this case study, no gravel sections are subjected to **high** traffic, so the gravel network is represented by two sections only, one for **medium** traffic and one for **low** traffic. The section ID (3 letters) is based on pavement type and traffic level as indicated below:

For GLT:

G	Gravel
LT	Light traffic

For GMT:

G	Gravel
MT	Medium traffic

The use of a matrix template to define the characteristics of the road network in terms of representative sections is discussed in Section 2.2 of Chapter B3. While this is useful in identifying groups of road sections with specific characteristics, the definition of properties for each representative section (for example, based on the average or weighted average of values within each group) is currently undertaken manually.

Double-clicking on a specific Section Description (or selecting the section and clicking the **View/Edit Section** button) reveals the **General/Geometry/Pavement/Condition** Tabs that give access to the section's details. Some relevant characteristics of the representative sections are summarised in Table C1.1and Table C1.2. Note that, for both bituminous and gravel sections, the condition must be defined at the start of the analysis period. As the analysis period commences in the year 2000, this means that the condition data must be defined for the year 1999 (that is, end of 1999) or earlier.

Table C1.1
Details of representative sections for bituminous roads

Section ID	**PHTGC**	**PHTFC**	**PHTPC**	**PMTGC**	**PMTFC**	**PMTPC**	**PLTGC**	**PLTFC**	**PLTPC**
Length (km)	234	392	437	306	483	615	410	670	720
Traffic level	High	High	High	Medium	Medium	Medium	Low	Low	Low
AADT	6200	5240	5180	2500	2300	2060	1400	1150	970
AADT year	1998	1998	1998	1998	1998	1998	1998	1998	1998
Condition (1999)	Good	Fair	Poor	Good	Fair	Poor	Good	Fair	Poor
Roughness (IRI)	2.5	4.0	6.5	2.5	4.4	6.5	3	4.5	6.1
Area of all cracks (%)	1	3	10	1	5	18	1	7	12
Ravelled area (%)	1	5	15	1	8	26	1	10	20
No of (small) potholes per km	0	3	20	0	5	40	0	8	60
Edge break (m^2/km)	0	5	10	0	5	15	2	12	30
Mean rut depth (mm)	2	5	10	2	5	12	3	10	15
Texture depth (mm)	3	2	0.5	3	2	0.5	3	2	0.5
Skid resistance (SCRIM 50 kph)	0.7	0.5	0.4	0.7	0.5	0.4	0.7	0.5	0.3
Drainage	Excellent	Fair	Poor	Excellent	Good	Fair	Good	Fair	Poor
Pavement type	AMAP	AMAP	AMAP	STGB	STGB	STGB	STGB	STGB	STGB
Surfacing material type	AC	AC	AC	DBSD	DBSD	AC	SBSD	SBSD	SBSD
New surf thick (mm)	50	50	50	25	25	50	15	15	15

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Section ID	PHTGC	PHTFC	PHTPC	PMTGC	PMTFC	PMTPC	PLTGC	PLTFC	PLTPC
Old surf thick (mm)	50	50	50	50	50	0	0	0	0
Date of last works types:									
- reconstruction	1991	1991	1991	1991	1991	1991	1991	1991	1991
- rehabilitation (overlay)	1996	1996	1996	1991	1991	1991	1991	1991	1991
- resurfacing (reseal)	1996	1996	1996	1996	1996	1991	1991	1991	1991
- preventative treatment	1996	1996	1996	1996	1996	1991	1991	1991	1991
SN after last treatment	2.70	2.70	2.70	2.07	2.07	2.26	1.67	1.67	1.67

Table C1.2
Details of representative sections for gravel roads

Section ID	**GMT**	**GLT**
Length (km)	1385	1760
Traffic level	Medium	Low
AADT (1998)	175	75
Carriageway width (m)	6	6
Shoulder width (m)	1	1
Condition (1998)		
Gravel thickness (mm)	100	70
Roughness IRI (m/km)	9	12

A tabulated data summary is useful in checking consistency of data across sections and also within a section. This includes checking that the condition classification is compatible with the condition parameters, and, with bituminous pavements, that the pavement type (defined under **Section/General**) is compatible with the pavement details (surfacing material type, surfacing thicknesses and dates of previous works) defined under **Section/Pavement**.

Remember that the dates of previous works refer to HDM-4 Works Types. Hence the **date of last resurfacing** refers to Works Type **resurfacing** which includes works activities such as surface dressing and slurry seal. Works activities such as overlaying, mill and replace, and inlay are classed as Works Type **rehabilitation** and would be recorded under **date of last rehabilitation**. The bituminous pavement types are reviewed below.

Bituminous pavement types

Table C1.1 indicates Pavement Type AMAP for the bituminous sections carrying high traffic levels (PHTGC, PHTFC and PHTPC). This is based on an original construction of 50 mm AC on a granular roadbase (that is, pavement type AMGB) in 1991 (date of last reconstruction), subsequently overlaid with 50 mm AC in 1996 (date of last rehabilitation). This results in the updated Pavement Type AMAP, and the dates of last resurfacing and last preventative treatment are set to 1996 (equal to date of last rehabilitation). Note the most recent surfacing thickness set at 50 mm (that is, overlay) and previous thickness is also 50 mm (original construction).

Pavement Type STAP is indicated for the sections identified as PMTGC and PMTFC. This is based on an original construction of 50 mm AC on a granular roadbase (pavement type AMGB) in 1991 (date of last reconstruction), subsequently surface dressed in 1996 (date of last resurfacing). This gives the updated pavement type STAP. The date of the last rehabilitation is set to 1991 (equal to date of original construction), and the **date of last preventative treatment** is set to 1996 (equal to date of last resurfacing). Note the most recent surfacing thickness is 25 mm (double surface dressing), and previous surfacing thickness is 50 mm (original construction).

The road section identified as **PMTPC** has pavement type AMGB that represents the original construction of 50 mm AC on a granular roadbase in 1991 (date of last reconstruction). Note that the dates of last rehabilitation, resurfacing and preventative treatment are also set to 1991 (equal to date of last reconstruction). In this case the most recent surfacing thickness is set to 50 mm (original construction), while the previous surfacing thickness is zero.

The three sections with low traffic level (PLTGC, PLTFC and PLTPC) have pavement type STGB, representing the original construction of double surface dressing on a granular roadbase in 1991 (date of last reconstruction). The dates of last rehabilitation, resurfacing and preventative treatment are also set at 1991. The most recent surfacing thickness is set to 25 mm (double surface dressing), while the previous surfacing thickness is zero.

Note that the Adjusted Structural Number (SN or SNP as appropriate) assigned to each section represents the condition of the pavement immediately after the most recent treatment (that is, date of last preventative treatment as defined under the **Pavement** Tab). For example, the SN value assigned to section PHTGC above represents Pavement Type AMAP (specified under the **Definition** Tab) immediately after the overlay application in 1996. The value of SN (surfacing, roadbase and sub-base only) can be quickly approximated as indicated below, based on the original AASHTO relationship:

$$SN = 0.0394 \sum a_i h_i \qquad ...(1.1)$$

where:

a_i layer coefficient

h_i layer thickness (mm)

Note the reduced value of layer coefficient (0.2) applied to the old surfacing due to its condition.

Table C1.3
Layer coefficients and thicknesses

Construction date	Layer	Layer thickness (mm) h_i	Layer coefficient a_i	$a_i h_i$
1996	AC Overlay	50	0.4	20
1991	AC	50	0.2	10
1991	Granular	150	0.14	21
1991	Granular	150	0.11	16.5
			Total	67.5

Giving:

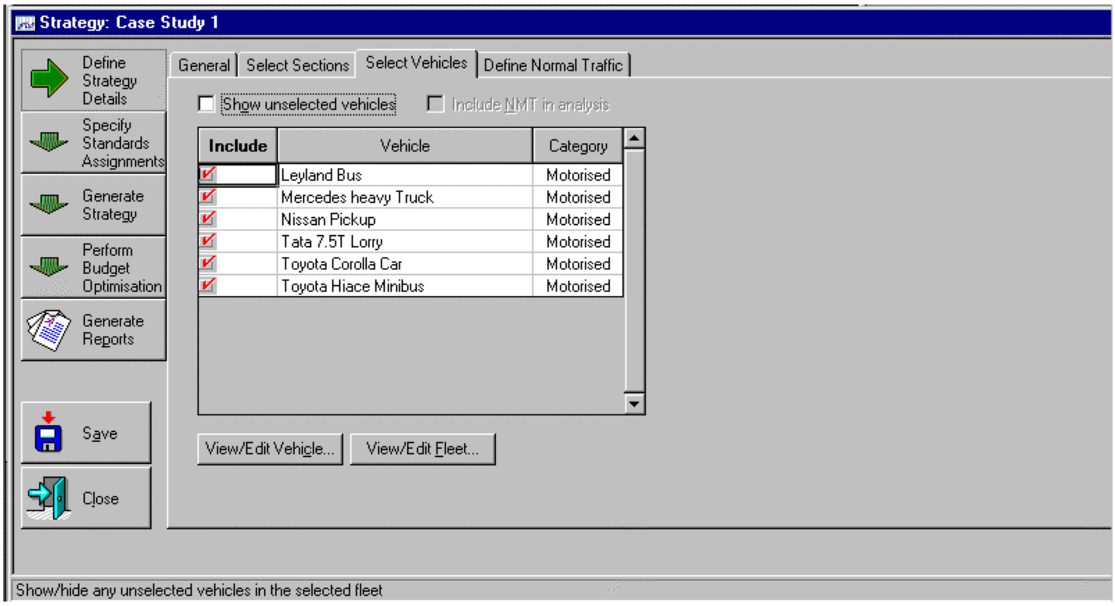
$$SN \cong 0.0394 * 67.5 = 2.7 \qquad ...(1.2)$$

This value has been assigned to section PHTGC under the **Section/Pavement** Tab.

The subgrade contribution (added to SN to give SNP) is computed by the HDM-4 software.

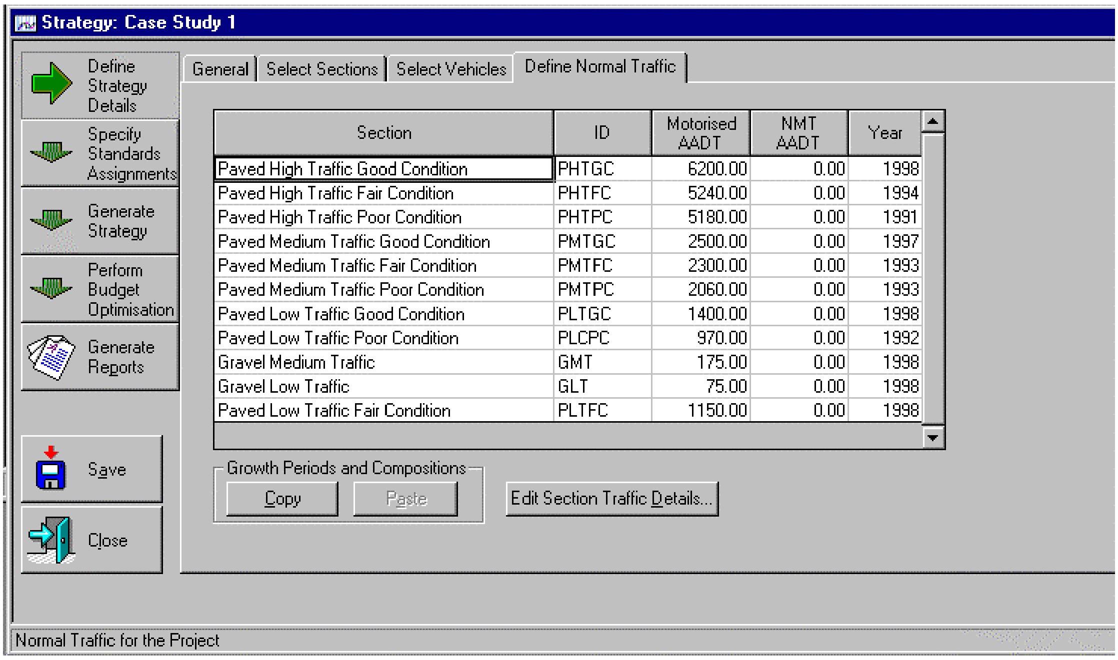
Note that this approximation does not take account of the reduced contribution from the sub-base, but is a much quicker manual calculation. Details of HDM-4 Structural Number concepts are given in Chapter C1 of the Analytical Framework and Model Descriptions.

Select Vehicles



The vehicles selected from the pre-defined **Western Province** fleet are indicated above. Individual vehicle attributes may be reviewed by double-clicking on the appropriate vehicle type description (or by selecting the section and clicking the **Edit Vehicle** button).

Define Normal Traffic



This screen confirms the traffic volume (AADT) on the selected road section in the given year. The initial traffic composition and growth rates (by vehicle type) can be reviewed by double-clicking the section description (or by *selecting* the section and clicking the **Section/Traffic/Details** button). The initial vehicle compositions are summarised in Table C1.4.

Note that the initial composition is defined for same year as the AADT. The annual growth rates may be specified for several periods, but must at least be specified for a period commencing from the analysis period start year (that is, 2000 in this case study). For example the AADT for section PHTGC is specified for 1998, and the initial vehicle composition applies to that year. The annual growth rates have been defined only for the period commencing 2000, and in this case are also applied between 1998 and 2000.

Table C1.4
Details of initial vehicle composition and growth rates

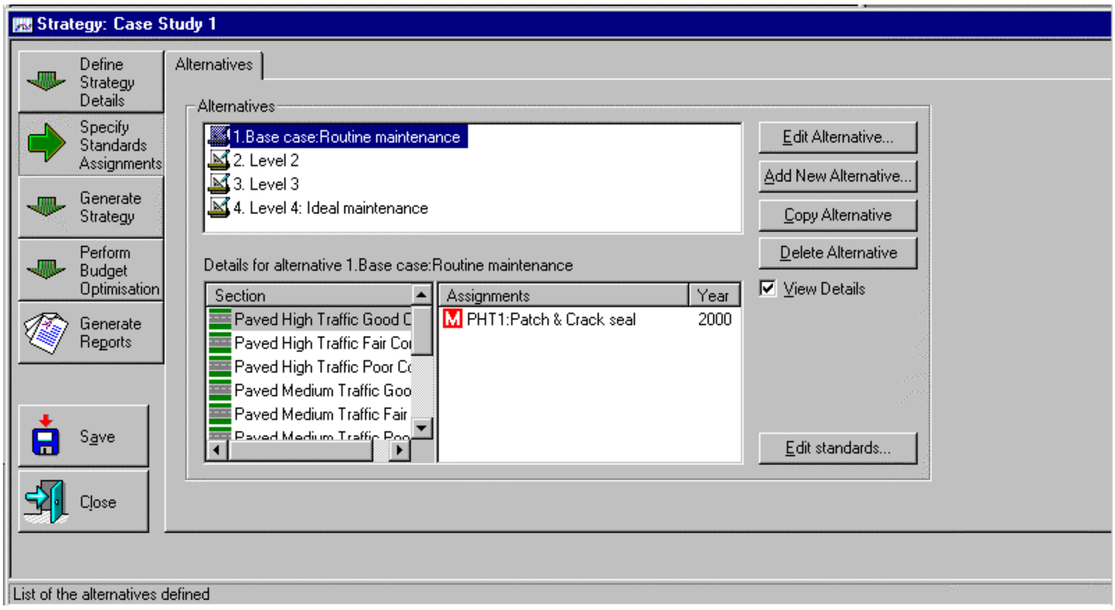
	Traffic level					
	High		Medium		Low	
Vehicle type	Initial composition %	Annual growth rate %	Initial composition %	Annual growth rate %	Initial composition %	Annual growth rate %
Leyland bus	10	3	10	3	0	0
Mercedes heavy truck	10	3	25	3	10	3
Nissan pickup	15	3	20	3	20	3
Tata 7.5t lorry	15	3	0	0	10	3
Toyota Corolla car	40	3	30	3	40	3
Toyota Hiace minibus	10	3	15	3	20	3
Total (%)	100		100		100	

Note: Traffic details (AADT & associated year, initial composition and associated growth rates) are assigned by section.

1.2.2 Specify Standards Assignments

For Strategy Analysis, the assignment of works is identical to that of Project Analysis. The assigned standards can be checked under:

Specify Standards Assignments/Alternatives



By clicking the **View Details** box, the user may review the assignment of Road Works Standards by Alternative and Section.

Alternatives

For this case study, four investment alternatives are considered for each of the representative sections, ranging from the provision of routine pavement maintenance only (Alternative 1) to an **ideal maintenance** case (Alternative 4). For each investment alternative, Road Works Standards have been pre-defined as shown in Table C1.5 and Table C1.6.

Table C1.5
Assigned Road Works Standards - short codes

Investment alternative		High traffic	Medium traffic	Low traffic
Bituminous (paved)				
1.	Patch & Crack seal	PHT1	PMT1	PLT1
2.	Patch, Reseal & Reconstruct	PHT2	PMT2	PLT2
3.	Patch, Overlay & Reconstruct	PHT3	PMT3	PLT3
4.	Patch, Reseal, Overlay & Reconstruct	PHT4	PMT4	PLT4
Gravel (unpaved)				
1.	Grade & Spot regravel		G1	G1
2.	Grade, Spot regravel & Regravel		G2	G2
3.	Grade, Spot regravel & Regravel		G3	G3
4.	Maintenance & Improvement		G1/GU4/PLT3	G1/GW4/G2

With bituminous road sections, maintenance standards have been assigned by alternative and traffic class. For gravel road sections, a maintenance standard has been defined for each investment alternative, and an improvement standard has been introduced to Alternative 4 (that is, upgrading gravel roads with medium traffic followed by paved road maintenance, and widening gravel roads with low traffic).

(Note that Road Authorities may wish to assign standards by Road Class, which may be represented in some countries by Surface Class alone. With condition responsive maintenance intervention, the interval between maintenance applications will clearly be influenced by the traffic volume.)

For **bituminous** (paved) sections, the Maintenance Standard short code is derived from the pavement surfacing, traffic class, and investment alternative, for example:

PHT1 is based on:

P	Paved
HT	High Traffic
1	Alternative 1

For **gravel** sections, the short code for Maintenance Standards is derived from the pavement surfacing material and the investment alternative:

For example:

G1	represents the Maintenance Standard for a gravel section under Alternative 1
GU4	represents the Improvement Standard to upgrade under Alternative 4
GW4	represents the Improvement Standard to widen under Alternative 4

The definition of investment alternatives for each of the representative sections is indicated in Table C1.6. This shows the short code identifier associated with each of the pre-defined Road Works Standards (that is, Maintenance Standards and Improvement Standards).

Table C1.6
Definition of investment alternatives for representative sections

Represent-ative section	Surface class	Traffic class	Investment alternative	Maintenance standard	Improve-ment standard	Future maint-enance
PHTGC	B	H	1	PHT1	-	-
			2	PHT2	-	-
			3	PHT3	-	-
			4	PHT4	-	-
PHTFC	B	H	1	PHT1	-	-
			2	PHT2	-	-
			3	PHT3	-	-
			4	PHT4	-	
PHTPC	B	H	1	PHT1	-	-
			2	PHT2	-	-
			3	PHT3	-	-
			4	PHT4	-	-
PMTGC	B	M	1	PMT1	-	-
			2	PMT2	-	-
			3	PMT3	-	-
			4	PMT4	-	-
PMTFC	B	M	1	PMT1	-	-
			2	PMT2	-	-
			3	PMT3	-	-
			4	PMT4	-	-
PMTPC	B	M	1	PMT1	-	-
			2	PMT2	-	-
			3	PMT3	-	-
			4	PMT4	-	-
PLTGC	B	L	1	PLT1	-	-
			2	PLT2	-	-
			3	PLT3	-	-
			4	PLT4	-	-

…Continued

Represent-ative section	Surface class	Traffic class	Investment alternative	Maintenance standard	Improve-ment standard	Future maint-enance
PLTFC	B	L	1	PLT1	-	
			2	PLT2	-	
			3	PLT3	-	-
			4	PLT4	-	-
PLTPC	B	L	1	PLT1	-	-
			2	PLT2	-	-
			3	PLT3	-	-
			4	PLT4	-	-
GMT	U	M	1	G1	-	-
			2	G2	-	-
			3	G3	-	-
			4	G1	GU4	PLT3
GLT	U	L	1	G1	-	-
			2	G2	-	-
			3	G3	-	-
			4	G1	GW4	G2

Note: Details of Road Works Standards identified by the above short codes are given in Table C1.9 and Table C1.11 for bituminous and gravel sections respectively.

Maintenance Standards for bituminous sections

The Maintenance Standards for **bituminous** sections are reviewed first. These are based on different combinations of the following works activities/operations.

Table C1.7
Works activities considered for bituminous sections

Works type	Works activity	Ranking
Routine Pavement	Patching	22
	Crack sealing	22
Resurfacing	Surface dressing double	16
Rehabilitation	Overlay dense graded asphalt	9
Reconstruction	Pavement reconstruction	6

If more than one of these activities is triggered in a given year, the works item that is highest in the hierarchy of road works (that is, with lowest ranking value) will be applied. Note that the routine pavement works (patching and crack sealing) have the same ranking, and both can be applied in the same maintenance year. (The hierarchy is defined by the ranking of road works shown in Chapter D2 of the Analytical Framework and Model Descriptions.

- **Alternative 1**

  Comprises patching and crack sealing only. This represents the base case (or **do minimum)** alternative. In this case study the intervention levels for patching vary from 10 potholes/km for sections with high traffic, to 50 potholes/km for sections with low traffic (see Table C1.9).

- **Alternative 2**

  Includes patching, resealing and reconstruction. The objective of this alternative is to adopt relatively inexpensive treatments which will maintain the existing road in a reasonable condition for as long as possible until the eventual need for reconstruction.

- **Alternative 3**

  Includes patching, overlaying and reconstruction. This alternative is effectively Alternative 2 with resealing replaced by overlaying. Alternative 3 introduces rehabilitation works at a pre-defined roughness level which will maintain a higher serviceability level than Alternative 2, as the structural strength of the pavement will increase at each overlay application.

- **Alternative 4**

  Includes patching, resealing, overlaying and reconstruction. This alternative combines the benefits of resealing and overlaying within one standard, which should reduce the required frequency of the relatively expensive overlay works.

The Maintenance Standards for alternatives 2, 3 and 4 include reconstruction as a Works Item. Details of the pavement layer thicknesses adopted for different road classes are given in Table C1.8. The pavement details will normally be derived using appropriate design charts, based on a design subgrade CBR and estimated cumulative traffic loading over the desired design life. The subgrade CBR value is not requested within the reconstruction works item, as this is already specified at section level (Section/Pavement).

Table C1.8
Pavement reconstruction details for bituminous sections

Details	Traffic class			Where specified in software
	Light	Medium	Heavy	
Pavement Type	STGB	STGB	AMGB	M/Design
Surface material	DBSD	DBSD	AC	M/Design
Surface thickness (mm)	25	25	50	M/Design
Structural Number (SN)	1.8	2.1	2.9	M/Design
Subgrade CBR (%)	8	8	8	Section/Pavement

Note: M/Design refers to the **Design** Tab under the appropriate Maintenance Standard

Details of the Maintenance Standards associated with each alternative are summarised in Table C1.9. The details may be reviewed under **Maintenance Standards** in HDM-4 Workspace or under **Specify Standards Assignments/Alternatives/Edit Standards**.

Table C1.9
Details of investment alternatives for bituminous sections

Investment alternative	Maintenance Standard Works item	Condition responsive criteria	Units	Intervention levels by traffic class		
				High traffic	Medium traffic	Low traffic
1	**Patch & crack seal**			**(PHT1)**	**(PMT1)**	**(PLT1)**
	Pothole patching	Potholing	no/km	>=10	>=25	>=50
	Crack sealing	Transverse thermal cracks AND	no/km	>=15	>=15	>=15
		Wide structural cracking	%	>=10	>=10	>=10
2	**Patch, reseal & reconstruct**			**(PHT2)**	**(PMT2)**	**(PLT2)**
	Pothole patching	Potholing	no/km	>=10	>=25	>50
	Reseal (SBSD)	Total damaged area	%	>=20	>=30	>40
	Reconstruct	Roughness	m/km IRI	>=10	>=11	>12
3	**Patch, overlay & reconstruct**			**(PHT3)**	**(PMT3)**	**(PLT3)**
	Pothole patching	Potholing	no/km	>=10	>=25	>=50
	Overlay	Roughness AND	m/km IRI	>=5	>=5.5	>=6
		Cracking	%	>=5	>=5	>=5
	Reconstruct	Roughness	m/km IRI	>=10	>11	>=12

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				Intervention levels by traffic class		
4	**Patch, reseal, overlay & reconstruct**			**(PHT4)**	**(PMT4)**	**(PLT4)**
	Pothole patching	Potholing	no/km	>=10	>=25	>=50
	Reseal (SBSD)	Total damaged area	%	>=20	>=30	>=40
	Overlay	Roughness AND	m/km IRI	>=5	>=5.5	>=6
		Cracking	%	>=5	>=5	>=5
	Reconstruct	Roughness	m/km IRI	>=10	>=11	>=12

Notes:

1 Short code identifiers for Maintenance Standards shown in parentheses, for example (PHT1)

2 SBSD = Single Bitumen Surface Dressing

Maintenance and Improvement Standards for gravel sections

The Maintenance and Improvement Standards for **gravel** sections are based on combinations of the works activities listed in Table C1.10.

Table C1.10
Works activities considered for gravel sections

M/I	Works type	Works activity	Ranking
M	Routine pavement	Grading	7
		Spot regravelling	7
M	Resurfacing	Regravelling	6
I	Widening	Partial widening	5
		Lane addition	4
I	Upgrading	Upgrading to a new surface class	2

The order of ranking of the Maintenance works items gives regravelling highest priority (that is, with lowest ranking value). Spot regravelling and Grading have equal ranking, and both of these can be performed in a given analysis year. Details of precedence rules for road works are given in Chapter D2 of the Analytical Framework and Model Descriptions.

- **Alternative 1**

  Comprises grading and spot regravelling, both condition responsive. This represents the base case (**do-minimum** alternative).

- **Alternative 2**

  Includes three works items, grading, spot regravelling and regravelling (resurfacing), all condition responsive. Grading is triggered by roughness, while spot regravelling and regravelling are both triggered by the thickness of gravel surfacing. This alternative aims to maintain a reasonable thickness of gravel surfacing and so provide continuous protection to the pavement subgrade.

- **Alternative 3**

  Includes the same works items as Alternative 2, but the intervention levels are set to provide a higher level of serviceability that should trigger the works more frequently.

- **Alternative 4**

  Introduces upgrading to a paved standard for sections with medium traffic (representative section GMT), and widening for sections with low traffic (section GLT). This represents the ideal investment alternative. Suitable maintenance is provided before and after the respective improvements. Note that the respective Improvement Standards are effective from the year 2003, with works scheduled to start in the year 2004. This means that no maintenance will be applied during the year 2003, the year prior to improvement.

Details of the Road Works Standards for gravel road sections are given in Table C1.11. The details can be reviewed under **Maintenance Standards** in HDM-4 Workspace or under:

Specify Standards Assignments/Alternatives/Edit Standards

Table C1.11
Details of investment alternatives for gravel sections

Alt	Sections	M / I	Road Works Standard Works Item	Effective from year	Responsive Criteria [M] Start Date/Duration [I]	Details (design)
1	GMT & GLT	M	**G1: Grade & Spot regravel**	2000		
			Grading		Roughness >= 8 IRI	
			Spot regravelling		GThk <= 100 mm	Replace 10% annual material loss
2	GMT & GLT	M	**G2: Grade , Spot regravel and Regravel**	2000		
			Grading		Roughness >= 8 IRI	
			Spot regravelling		GThk <= 100 mm	Replace 10% annual material loss
			Regravelling		GThk <= 50 mm	Final gravel thickness 150 mm
3	GMT & GLT	M	**G3: Grade, Spot regravel and Regravel**	2000		
			Grading		Roughness >= 7 IRI	
			Spot regravelling		GThk <= 125 mm	Replace 10% annual material loss
			Regravelling		GThk <= 75 mm	Final gravel thickness 150 mm

…Continued

Alt	Sections	M / I	Road Works Standard Works Item	Effective from year	Responsive Criteria [M] Start Date/Duration [I]	Details (design)
4	GMT	M	**G1 : Grade and Spot regravel** (details above)	2000		
		I	**GU4: Pave section GMT in 2004**	2003	Start year 2004 Duration 5 years	Upgrade to STGB with 2 m widening New Speed-Flow Type is 2-lane wide
		M	**PLT3: Patch, Overlay & Reconstruct**	2009		
			Pothole patching		Potholing >= 50/km	
			Overlay		Roughness >= 6 IRI AND Cracking >= 5%	
			Reconstruct		Roughness >= 12 IRI	
5	GLT	M	**G1 : Grade and Spot regravel** (details above)	2000		
		I	**GW4: Widen section GLT in 2004**	2003	Start year 2004 Duration 5 years	Widen by 3 m New Speed-Flow Type is 2-lane wide
		M	**G2: Grade, Spot regravel and Regravel** (details above)	2009		

Notes:

Column 1 Alt = Investment Alternative

Column 3 M/I = Maintenance or Improvement Standard

Column 6 GThk = Gravel thickness

1.3 Funding requirements without budget constraint

This step in Strategy Analysis produces an unconstrained work programme, based on a life-cycle analysis that considers the different alternatives for each representative section. The alternative giving the highest NPV is assigned to each section.

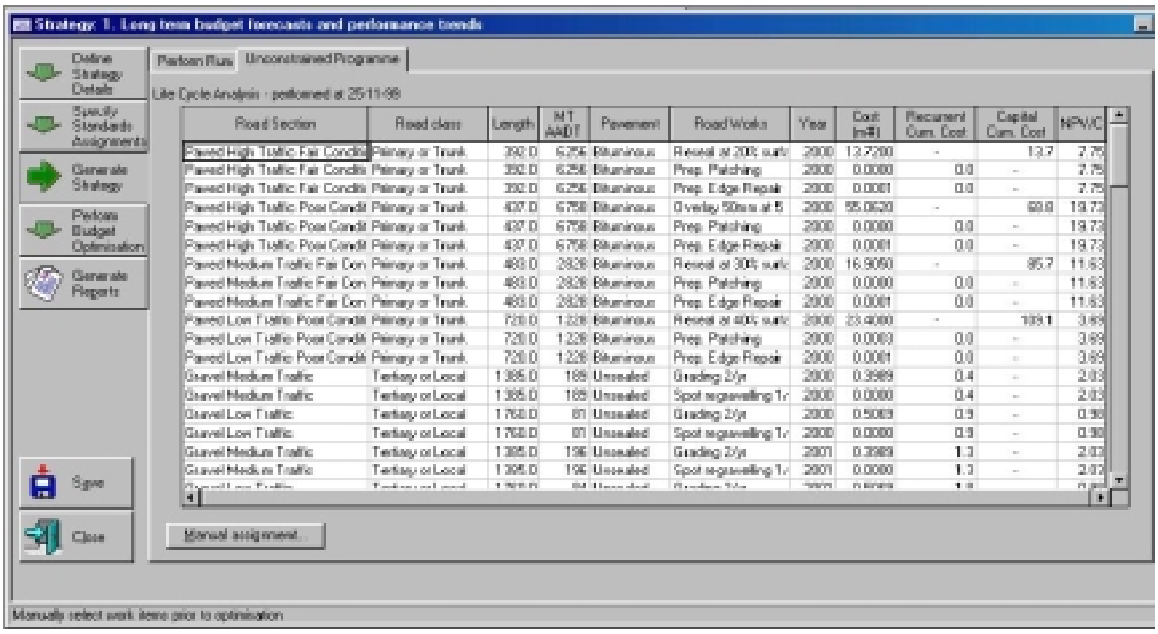
1.3.1 Generate Strategy

Perform Run / Run Setup

The Run Setup screen for Strategy Analysis is identical to that for life-cycle analysis in Programme Analysis. The **multi-year forward programme** option (available under Programme Analysis) is not applicable to strategy analysis that is concerned with longer term planning.

Work Programme

On completing Run Setup, press the **Start** button to commence the economic analysis, which produces a costed work programme containing, for each representative section, the alternative with the highest NPV (refer to Section 3.2.3 of Chapter B2). This represents an unconstrained programme, with the total financial cost given in the **cumulative cost** column. The road works assigned to each section represent the optimum maintenance and improvements that should be applied to the road network in accordance with the specified standards.



Road Section	Road class	Length	MT AADT	Pavement	Road Works	Year	Cost (m$)	Recurrent Cum. Cost	Capital Cum. Cost	NPV/C
Paved High Traffic Fair Conditi	Primary or Trunk	392.0	6256	Bituminous	Reseal at 20% surfa	2000	13.7200	-	13.7	7.75
Paved High Traffic Fair Conditi	Primary or Trunk	392.0	6256	Bituminous	Prep. Patching	2000	0.0000	0.0	-	7.75
Paved High Traffic Fair Conditi	Primary or Trunk	392.0	6256	Bituminous	Prep. Edge Repair	2000	0.0001	0.0	-	7.75
Paved High Traffic Poor Condit	Primary or Trunk	437.0	6758	Bituminous	Overlay 50mm at 5	2000	55.0620	-	68.8	19.73
Paved High Traffic Poor Condit	Primary or Trunk	437.0	6758	Bituminous	Prep. Patching	2000	0.0000	0.0	-	19.73
Paved High Traffic Poor Condit	Primary or Trunk	437.0	6758	Bituminous	Prep. Edge Repair	2000	0.0001	0.0	-	19.73
Paved Medium Traffic Fair Con	Primary or Trunk	483.0	2828	Bituminous	Reseal at 30% surfa	2000	16.9050	-	85.7	11.63
Paved Medium Traffic Fair Con	Primary or Trunk	483.0	2828	Bituminous	Prep. Patching	2000	0.0000	0.0	-	11.63
Paved Medium Traffic Fair Con	Primary or Trunk	483.0	2828	Bituminous	Prep. Edge Repair	2000	0.0001	0.0	-	11.63
Paved Low Traffic Poor Conditi	Primary or Trunk	720.0	1228	Bituminous	Reseal at 40% surfa	2000	23.4000	-	109.1	3.69
Paved Low Traffic Poor Conditi	Primary or Trunk	720.0	1228	Bituminous	Prep. Patching	2000	0.0003	0.0	-	3.69
Paved Low Traffic Poor Conditi	Primary or Trunk	720.0	1228	Bituminous	Prep. Edge Repair	2000	0.0001	0.0	-	3.69
Gravel Medium Traffic	Tertiary or Local	1385.0	189	Unsealed	Grading 2/yr	2000	0.3989	0.4	-	2.03
Gravel Medium Traffic	Tertiary or Local	1385.0	189	Unsealed	Spot regravelling 1/	2000	0.0000	0.4	-	2.03
Gravel Low Traffic	Tertiary or Local	1760.0	81	Unsealed	Grading 2/yr	2000	0.5069	0.9	-	0.90
Gravel Low Traffic	Tertiary or Local	1760.0	81	Unsealed	Spot regravelling 1/	2000	0.0000	0.9	-	0.90
Gravel Medium Traffic	Tertiary or Local	1385.0	196	Unsealed	Grading 2/yr	2001	0.3989	1.3	-	2.03
Gravel Medium Traffic	Tertiary or Local	1385.0	196	Unsealed	Spot regravelling 1/	2001	0.0000	1.3	-	2.03

1.4 Effect of budget constraints

This step in Strategy Analysis is concerned with budget optimisation, whereby a budget is defined, and an optimisation procedure is selected which aims to produce a works programme that satisfies an objective function, that is:

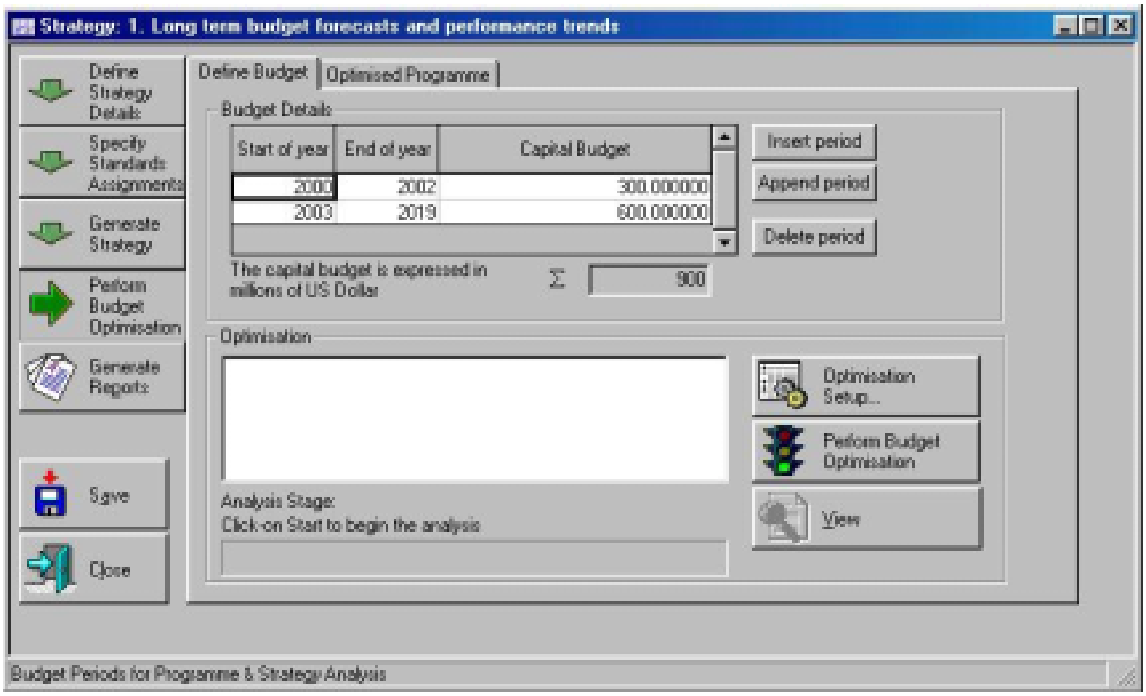
either maximisation of benefits (NPV)

or maximisation of the improvement in network condition

For this case study, two budget scenarios are considered, one at 75% of funding requirements (from Step 3), and another at 50%. For each of these, optimisation has been based on maximisation of benefits, and the long term performance trends predicted.

1.4.1 Perform budget optimisation (budget at 75% funding requirement)

Define Budget



The budget for the defined budget period has been set at 900 million US dollars.

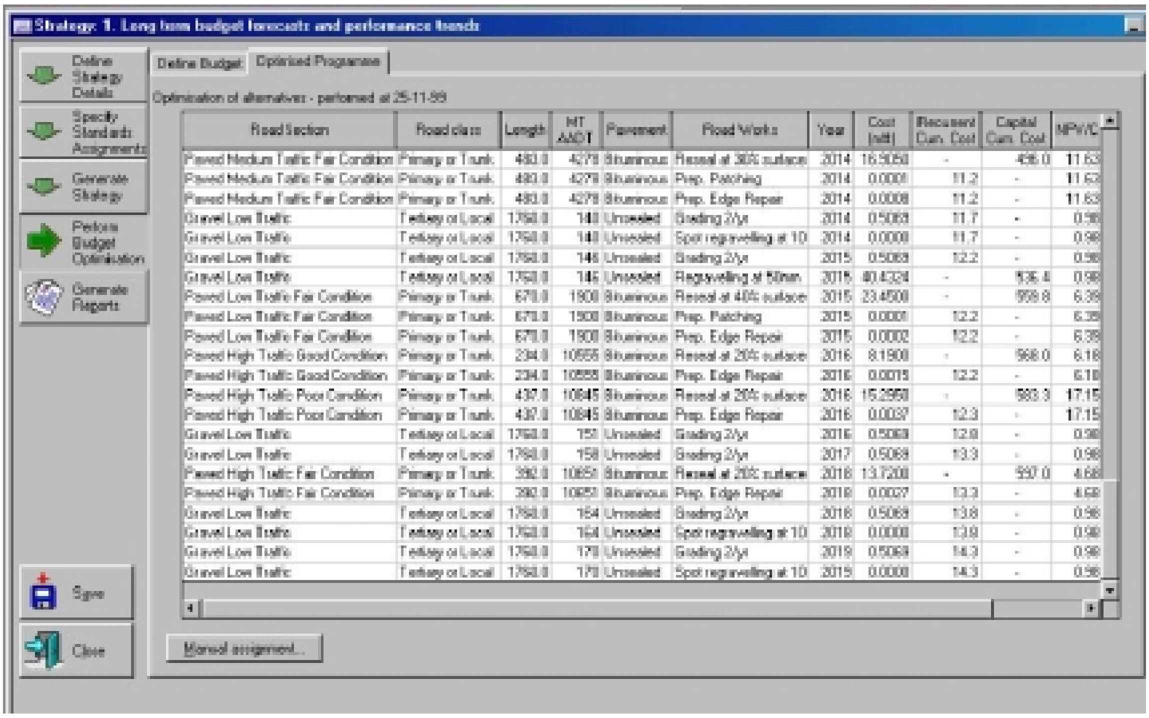
Details of the budget optimisation set-up are specified on this screen. For this case study, optimisation is by incremental benefit/cost ranking (see Section 4.2 of Chapter B2). The parameters to be specified are given below:

Minimum incremental value = 0

Efficiency frontier zone = 95

Work Programme

After performing the budget optimisation, a revised work programme is produced. This has a total cost that falls within the specified budget. The road sections are listed in priority order in accordance with the incremental benefit/cost ranking procedure described in Section 4.2 of Chapter B2.



Road Section	Road class	Length	MT AADT	Pavement	Road Works	Year	Cost (mill)	Recurrent Cum. Cost	Capital Cum. Cost	NPV/C
Paved Medium Traffic Fair Condition	Primary or Trunk	480.0	4278	Bituminous	Reseal at 30% surface	2014	16.5050	-	496.0	11.63
Paved Medium Traffic Fair Condition	Primary or Trunk	480.0	4278	Bituminous	Prep. Patching	2014	0.0001	11.2	-	11.63
Paved Medium Traffic Fair Condition	Primary or Trunk	480.0	4278	Bituminous	Prep. Edge Repair	2014	0.0008	11.2	-	11.63
Gravel Low Traffic	Tertiary or Local	1760.0	140	Unsealed	Grading 2/yr	2014	0.5069	11.7	-	0.98
Gravel Low Traffic	Tertiary or Local	1760.0	140	Unsealed	Spot regravelling at 10	2014	0.0000	11.7	-	0.98
Gravel Low Traffic	Tertiary or Local	1760.0	146	Unsealed	Grading 2/yr	2015	0.5069	12.2	-	0.98
Gravel Low Traffic	Tertiary or Local	1760.0	146	Unsealed	Regravelling at 50mm	2015	40.4324	-	536.4	0.98
Paved Low Traffic Fair Condition	Primary or Trunk	670.0	1900	Bituminous	Reseal at 40% surface	2015	23.4500	-	959.8	6.39
Paved Low Traffic Fair Condition	Primary or Trunk	670.0	1900	Bituminous	Prep. Patching	2015	0.0001	12.2	-	6.39
Paved Low Traffic Fair Condition	Primary or Trunk	670.0	1900	Bituminous	Prep. Edge Repair	2015	0.0002	12.2	-	6.39
Paved High Traffic Good Condition	Primary or Trunk	234.0	10555	Bituminous	Reseal at 20% surface	2016	8.1900	-	968.0	6.18
Paved High Traffic Good Condition	Primary or Trunk	234.0	10555	Bituminous	Prep. Edge Repair	2016	0.0015	12.2	-	6.18
Paved High Traffic Poor Condition	Primary or Trunk	437.0	10845	Bituminous	Reseal at 20% surface	2016	15.2950	-	983.3	17.15
Paved High Traffic Poor Condition	Primary or Trunk	437.0	10845	Bituminous	Prep. Edge Repair	2016	0.0037	12.3	-	17.15
Gravel Low Traffic	Tertiary or Local	1760.0	151	Unsealed	Grading 2/yr	2016	0.5069	12.8	-	0.98
Gravel Low Traffic	Tertiary or Local	1760.0	158	Unsealed	Grading 2/yr	2017	0.5069	13.3	-	0.98
Paved High Traffic Fair Condition	Primary or Trunk	390.0	10851	Bituminous	Reseal at 20% surface	2018	13.7200	-	997.0	4.68
Paved High Traffic Fair Condition	Primary or Trunk	390.0	10851	Bituminous	Prep. Edge Repair	2018	0.0027	13.3	-	4.68
Gravel Low Traffic	Tertiary or Local	1760.0	164	Unsealed	Grading 2/yr	2018	0.5069	13.8	-	0.98
Gravel Low Traffic	Tertiary or Local	1760.0	164	Unsealed	Spot regravelling at 10	2018	0.0000	13.8	-	0.98
Gravel Low Traffic	Tertiary or Local	1760.0	170	Unsealed	Grading 2/yr	2019	0.5069	14.3	-	0.98
Gravel Low Traffic	Tertiary or Local	1760.0	170	Unsealed	Spot regravelling at 10	2019	0.0000	14.3	-	0.98

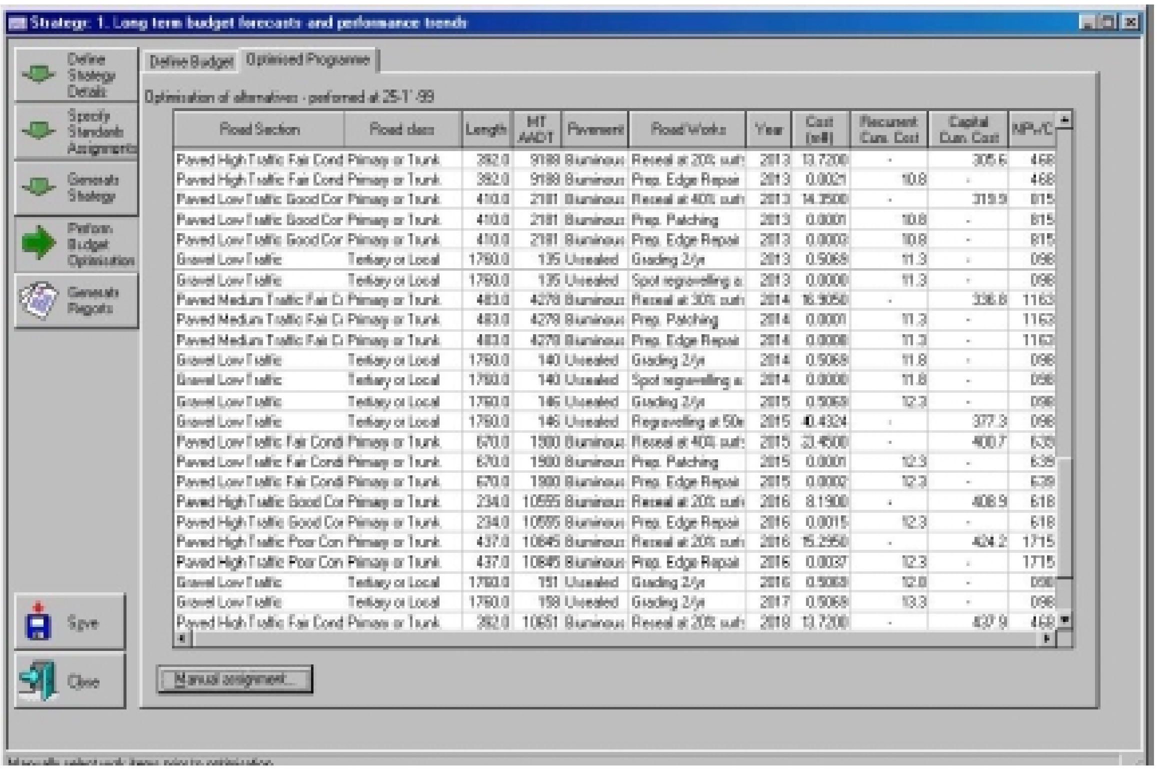
1.4.2 Perform Budget Optimisation (budget at 50% funding requirement)

Define Budget

The budget for the defined budget period has been set at 450 million US dollars, representing approximately 50% of the funding requirements from Step 3.

Work Programme

After performing the budget optimisation, the revised work programme is produced which shows the road sections listed in priority order in accordance with the incremental benefit/cost ranking procedure.



Road Section	Road class	Length	MT AADT	Pavement	Road Works	Year	Cost (m$)	Recurrent Cum. Cost	Capital Cum. Cost	NPV/C
Paved High Traffic Fair Cond	Primary or Trunk	392.0	9188	Bituminous	Reseal at 20% surf	2013	13.7200	-	305.6	468
Paved High Traffic Fair Cond	Primary or Trunk	392.0	9188	Bituminous	Prep. Edge Repair	2013	0.0021	10.8	-	468
Paved Low Traffic Good Cor	Primary or Trunk	410.0	2181	Bituminous	Reseal at 40% surf	2013	14.3500	-	319.9	815
Paved Low Traffic Good Cor	Primary or Trunk	410.0	2181	Bituminous	Prep. Patching	2013	0.0001	10.8	-	815
Paved Low Traffic Good Cor	Primary or Trunk	410.0	2181	Bituminous	Prep. Edge Repair	2013	0.0002	10.8	-	815
Gravel Low Traffic	Tertiary or Local	1780.0	135	Unsealed	Grading 2/yr	2013	0.5068	11.3	-	098
Gravel Low Traffic	Tertiary or Local	1780.0	135	Unsealed	Spot regravelling a	2013	0.0000	11.3	-	098
Paved Medium Traffic Fair Co	Primary or Trunk	483.0	4278	Bituminous	Reseal at 30% surf	2014	16.9050	-	336.8	1163
Paved Medium Traffic Fair Co	Primary or Trunk	483.0	4278	Bituminous	Prep. Patching	2014	0.0001	11.3	-	1163
Paved Medium Traffic Fair Co	Primary or Trunk	483.0	4278	Bituminous	Prep. Edge Repair	2014	0.0008	11.3	-	1163
Gravel Low Traffic	Tertiary or Local	1780.0	140	Unsealed	Grading 2/yr	2014	0.5068	11.8	-	098
Gravel Low Traffic	Tertiary or Local	1780.0	140	Unsealed	Spot regravelling a	2014	0.0000	11.8	-	098
Gravel Low Traffic	Tertiary or Local	1780.0	146	Unsealed	Grading 2/yr	2015	0.5068	12.3	-	098
Gravel Low Traffic	Tertiary or Local	1780.0	146	Unsealed	Regravelling at 50n	2015	40.4324	-	377.3	098
Paved Low Traffic Fair Condi	Primary or Trunk	670.0	1980	Bituminous	Reseal at 40% surf	2015	23.4500	-	400.7	639
Paved Low Traffic Fair Condi	Primary or Trunk	670.0	1980	Bituminous	Prep. Patching	2015	0.0001	12.3	-	639
Paved Low Traffic Fair Condi	Primary or Trunk	670.0	1980	Bituminous	Prep. Edge Repair	2015	0.0002	12.3	-	639
Paved High Traffic Good Cor	Primary or Trunk	234.0	10595	Bituminous	Reseal at 20% surf	2016	8.1900	-	408.9	618
Paved High Traffic Good Cor	Primary or Trunk	234.0	10595	Bituminous	Prep. Edge Repair	2016	0.0015	12.3	-	618
Paved High Traffic Poor Con	Primary or Trunk	437.0	10845	Bituminous	Reseal at 20% surf	2016	15.2950	-	424.2	1715
Paved High Traffic Poor Con	Primary or Trunk	437.0	10845	Bituminous	Prep. Edge Repair	2016	0.0037	12.3	-	1715
Gravel Low Traffic	Tertiary or Local	1780.0	151	Unsealed	Grading 2/yr	2016	0.5068	12.8	-	098
Gravel Low Traffic	Tertiary or Local	1780.0	158	Unsealed	Grading 2/yr	2017	0.5068	13.3	-	098
Paved High Traffic Fair Cond	Primary or Trunk	392.0	10651	Bituminous	Reseal at 20% surf	2018	13.7200	-	437.9	468